9

Inferences About Variances

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Key Concepts

1. Sampling distributions of the variance
2. The *F* distribution
3. Homogeneity of variance tests

In the previous three chapters we looked at testing inferences about means (Chapters 6 and 7) and about proportions (Chapter 8). In this chapter we examine inferential tests involving variances. Tests of variances are useful in two applications: (a) as an inferential test by itself and (b) as a test of the homogeneity of variance assumption for another procedure (e.g., *t* test, analysis of variance).

First, a researcher may want to perform inferential tests on variances for their own sake, in the same fashion that we described for the one- and two-sample *t* tests on means. For example, we may want to assess whether the variance of undergraduates at Ivy-Covered University on an intelligence measure is the same as the theoretically derived variance of 225 (from when the test was developed and normed). In other words, is the variance at a particular university greater than or less than 225? As another example, we may want to

determine whether the variances on an intelligence measure are consistent across two or more groups; for example, is the variance of the intelligence measure at Ivy-Covered University different from that at The Greatest University?

Second, for some procedures, such as the independent *t* test (Chapter 7) and the analysis of variance, it is assumed that the variances for two or more independent samples are equal (known as the homogeneity of variance assumption). Thus, we may want to use an inferential test of variances to assess whether this assumption has been violated or not. The following inferential tests of variance are covered in this chapter: (a) testing whether a single variance is different from a hypothesized value; (b) testing whether two dependent variances are different; and (c) testing whether two or more independent variances are different. We utilize many of the foundational concepts covered in Chapters 6, 7, and 8. New concepts to be discussed include the following: the sampling distributions of the variance, the *F* distribution, and homogeneity of variance tests. Our objectives are that by the end of this chapter, you will be able to (a) understand the basic concepts underlying tests of variances, (b) select the appropriate test, and (c) determine and interpret the results from the appropriate test.

9.1 Inferences About Variances and How They Work

As you remember, Oso Wyse is one of four extraordinarily talented graduate students who is working in the stats lab. Oso and colleagues have had the opportunity to work on quite a number of exciting statistical projects. We revisit the group again, with Oso getting ready to embark on another stats journey.

Another call has been fielded by the stats lab for assistance with statistical analysis. This time, it is Dr. Abraham, an elementary assistant principal within the community. Dr. Abraham shares with Oso that she is conducting a teacher research project related to achievement of first grade students at her school. Dr. Abraham wants to determine if the variances of the achievement scores differ when children begin school in the fall as compared to when they end school in the spring. Oso suggests the following research question: *Are the variances of achievement scores for first grade children the same in the fall as compared to the spring?* Oso suggests a test of variance as the test of inference. His task is then to assist Dr. Abraham in generating the test of inference to answer her research question.

This section deals with concepts for testing inferences about variances, in particular, the sampling distributions underlying such tests. Subsequent sections deal with several inferential tests of variances. Although the sampling distribution of the mean is a normal distribution (Chapters 6 and 7), and the sampling distribution of a proportion is either a normal or chi-square distribution (Chapter 8), the **sampling distribution of a variance** *is a chi-square distribution for a single variance, a* t *distribution for two dependent variances, or an* F *distribution for two or more independent variances*. Although we have already discussed the *t* distribution in Chapter 6 and the chi-square distribution in Chapter 8, we need to discuss the *F* distribution (named in honor of the famous statistician Sir Ronald A. Fisher) in some detail here.

The O'Brien procedure is an analysis of variance on a *different* transformed variable:

$$r_{ij} = \frac{(n_j - 1.5)(n_j)(Y_{ij} - \bar{Y}_j)^2 - (.5)(s_j^2)(n_j - 1)}{(n_j - 1)(n_j - 2)}$$

$$r_{ij} = \frac{(n_j - 1.5)n_j(Y_{ij} - \bar{Y}_j)^2 - .5s_j^2(n_j - 1)}{(n_j - 1)(n_j - 2)}$$

which is computed for each individual, where n_j is the size of group j, $\bar{Y}_{.j}$ is the mean on the outcome for group j, and s_j^2 is the sample variance for group j.

The test statistic is an F statistic and is computed by the following equation:

$$F = \frac{\sum_{j=1}^{J} n_j \left(\bar{r}_{.j} - \bar{r}_{..}\right)^2 \Big/ \left(J - 1\right)}{\sum_{i=1}^{n_j} \sum_{j=1}^{J} \left(r_{ij} - \bar{r}_{.j}\right)^2 \Big/ \left(N - J\right)}$$

where n_j designates the number of observations in group j, J is the number of groups (where j ranges from 1 to J), $\bar{r}_{.j}$ is the mean for group j (computed by taking the sum of the observations in group j and dividing by the number of observations in group j, which is n_j), and $r_{..}$ is the overall mean regardless of group membership (computed by taking the sum of all of the observations across all groups and dividing by the total number of observations N). The test statistic F is compared against a critical value from the F table (Table A.4 in the Appendix) with $J - 1$ degrees of freedom in the numerator and $N - J$ degrees of freedom in the denominator, denoted by ${}_{\alpha}F_{j-1,N-J}$ If the test statistic is greater than the critical value, then we reject H_0; otherwise, we fail to reject H_0.

Let us return to the example in Table 9.1 and consider the results of the O'Brien procedure. From the computations shown in the table, the test statistic $F = 1.4799$ is compared against the critical value for $\alpha = .05$ of ${}_{.05}F_{2,9} = 4.26$. Because the test statistic is smaller than the critical value (i.e., $1.4799 < 4.26$), we fail to reject the null hypothesis and conclude that the three student groups do not have different variances.

9.2 Assumptions

9.2.1 Assumptions for Inferences About a Single Variance

It is assumed that the sample is randomly drawn from the population (i.e., the assumption of independence) and that the population of scores is normally distributed. It has been noted by statisticians such as Wilcox (1996) that the chi-square distribution does not perform adequately when sampling from a nonnormal distribution, because the actual Type I error rate can differ greatly from the nominal alpha level (the level set by the researcher).

While not an assumption, because we are testing a variance, a condition of the test is that the variable must be *interval* or *ratio* in scale.

9.2.2 Assumptions for Inferences About Two Dependent Variances

It is assumed that the two samples are dependently and randomly drawn from their respective populations, that both populations are normal in shape, and that the *t* distribution is the appropriate sampling distribution. It is thought that this test is not particularly robust to nonnormality (Wilcox, 1987). As a result, other procedures have been developed that are thought to be more robust. However, little in the way of empirical results is known at this time.

While not an assumption, because we are testing a variance, a condition of the test is that the variable must be *interval* or *ratio* in scale. Recall that variances can only be computed with data that are interval or ratio in scale.

9.3 Sample Size, Power, and Effect Size

There is really not much we can report on that is available in published in the literature on sample size, power, and effect sizes for tests of variances.

9.4 Computing Inferences About Variances Using SPSS

Unfortunately, there is not much to report on tests of variances for SPSS. There are no unique (i.e., standalone) tests available for inferences about a single variance or for inferences about two dependent variances. For inferences about independent variances, SPSS does provide Levene's test as part of the "Independent *t* Test" procedure (discussed in Chapter 7) and as part of the "One Way ANOVA" and "Univariate ANOVA" procedures. While it is commonly reported as evidence for meeting the assumption of equal variances, given our previous concerns with Levene's test, use it with caution.

9.5 Computing Inferences About Variances Using R

Next we consider **R** for computing inferences about variances. We will examine both inferences about a single variance and inferences about two dependent variances.

Note that the scripts are provided within the blocks with additional annotation to assist in understanding how the command works. Should you want to write reminder notes and annotation to yourself as you write the commands in **R** (and we highly encourage doing so), remember that any text that follows a hashtag (i.e., #) is annotation only and not part of the **R** script. Thus, you can write annotations directly into **R** with hashtags. We encourage this practice so that when you call up the commands in the future, you'll understand what the various lines of code are doing. You may think you'll remember what you did. However, trust us. There is a good chance that you won't. Thus, consider it best practice when using **R** to annotate heavily!

9.5.4 Generating the Test of Inference About Two Dependent Variances

```
var(Ch9_swimming$pretest)
var(Ch9_swimming$posttest)
```

Before we compute the test for the inference about two dependent variances, let's generate the values of the variances for both the pretest and posttest. Using the *var* function for the two variables of interest, the variances are, respectively:

```
[1] 17.77778 #pretest
[1] 13.11111 #posttest
```

```
var.test(Ch9_swimming$pretest, Ch9_swimming$posttest,
        paired=TRUE,
        alternative = "two.sided",
        conf.level = 0.95)
```

We will use the *var.test* function to test the inference about two dependent variances. Let's review what is inside the parentheses. We first define the dataframe ("Ch9_swimming") and variables to compute the test ("Ch9_swimming$pretest" and "Ch9_swimming$posttest). This test is comparable to the dependent *t* test, comparing one population variance to another, and thus we define this as *paired=TRUE*. The *alternative* command specifies the alternative hypothesis the variance for the pretest is different than the variance for the posttest. Had we wanted to test a one-directional hypothesis, we would have had the command *alternative = "greater"* or *alternative = "less,"* respectively. We test at an alpha of .05, so the confidence level is .95 (*conf.level = 0.95*). Using this command, we are provided with the following results. We see the ratio of the variances is about 1.36. The results of the *F* test are not statistically significant ($F = 1.36$, $p = .68$).

```
Results of Hypothesis Test
--------------------------
Null Hypothesis:            ratio of variances = 1
Alternative Hypothesis:     True ratio of variances is not equal to 1
Test Name:                  F test to compare two variances
Estimated Parameter(s):     ratio of variances = 1.355932
Data:                       Ch9_swimming$pretest and Ch9_swimming$posttest
Test Statistic:             F = 1.355932
Test Statistic Parameters:  num df = 9
                            denom df = 9

P-value:                    0.6574637
95% Confidence Interval:    LCL = 0.3367944
                            UCL = 5.4589751
```

FIGURE 9.5
Generating a test of inference about two dependent variances.

9.6 Research Question Template and Example Write-Up

Consider an example paragraph for one of the tests described in this chapter, more specifically, testing inferences about two dependent variances. As you may remember, our graduate research assistant, Oso, was working with Dr. Abraham, an assistant principal, to assist in analyzing the variances of first grade students. Oso's task was to assist Dr. Abraham

with writing her research question (*Are the variances of achievement scores for first grade children the same in the fall as compared to the spring?*) and generating the test of inference to answer her question. Oso suggested a dependent variances test as the test of inference. A template for writing a research question for the dependent variances follows:

Are the variances of [variable] the same in [time 1] as compared to [time 2]?

The following is an example write-up:

A test of dependent variances was conducted to determine if variances of achievement scores for first grade children were the same in the fall as compared to the spring. The test was conducted using an alpha of .05. The null hypothesis was that the variances would be the same.

There was a statistically significant difference in variances of achievement scores of first grade children in the fall as compared to the spring ($t = -3.4261$, $df = 60$, $p < .05$). Thus the null hypothesis that the variances would be equal at the beginning and end of the first grade was rejected. The variances of achievement test scores significantly increased from September to April.

9.7 Additional Resources

We have offered a number of resources within the chapter and refer readers who are interested in learning more to those resources. Because homogeneity of variance is an integral assumption to tests of means, readers may also find coverage of tests of inference in texts that deal with ANOVA and related designs (e.g., Maxwell, Delaney, & Kelley, 2018).

Problems

Conceptual Problems

1. Which of the following tests of homogeneity of variance is most robust to assumption violations?
   a. *F* ratio test
   b. Bartlett's chi-square test
   c. O'Brien procedure
   d. Hartley's F_{max} test
2. True or false? Cochran's *C* test requires equal sample sizes.
3. I assert that if two dependent sample variances are identical, I would not be able to reject the null hypothesis. Am I correct?